

**Version: 1.0****TECHNICAL SPECIFICATION****MODEL NO: ET011TJ1**

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☐ Customer's Confirmation

Customer _____

Date _____

By _____

■ E Ink's ConfirmationApproved By 李纪博Confirmed By 吴琪铭Prepared By 吴岱融



Revision History

Rev.	Issued Date	Revised Contents
0.1	Jan. 18, 2017	Preliminary
0.2	Feb. 7, 2017	Wording modification of data polarity on Ch16-20 Update Ch4 Mechanical Drawing of EPD Module
0.3	July 17, 2017	Update Ch4 Mechanical Drawing of EPD Module Update Ch8 Optical Characteristics Update Ch17 Packing Update Ch5-2 & Ch6-5 Add Outline Dimension Add Module Weight
0.4	Aug. 11, 2017	Update Ch6 Electrical Characteristics Update Ch8 Optical Characteristics
0.5	Aug. 22, 2017	Update Ch6 Electrical Characteristics, to add preliminary data of Operating Temp. Range & Storage Temperature Update Ch10 Reliability Test, to add preliminary data of RA
1.0	Oct. 17, 2017	Ch6 Electrical Characteristics & Ch7 Power Sequence Format Changed Update Ch8 Optical Characteristics , White Chromaticity Update Ch10 Reliability Test



TECHNICAL SPECIFICATION

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1. General Description

ET011TJ1 is a reflective electrophoretic E Ink technology display module based on active matrix TFT and plastic substrate. It has 1.1" active area with 240 x 240 pixels, and the display is capable to display images at 2 gray levels (1 bit) depending on the display controller and the associated waveform file it used.

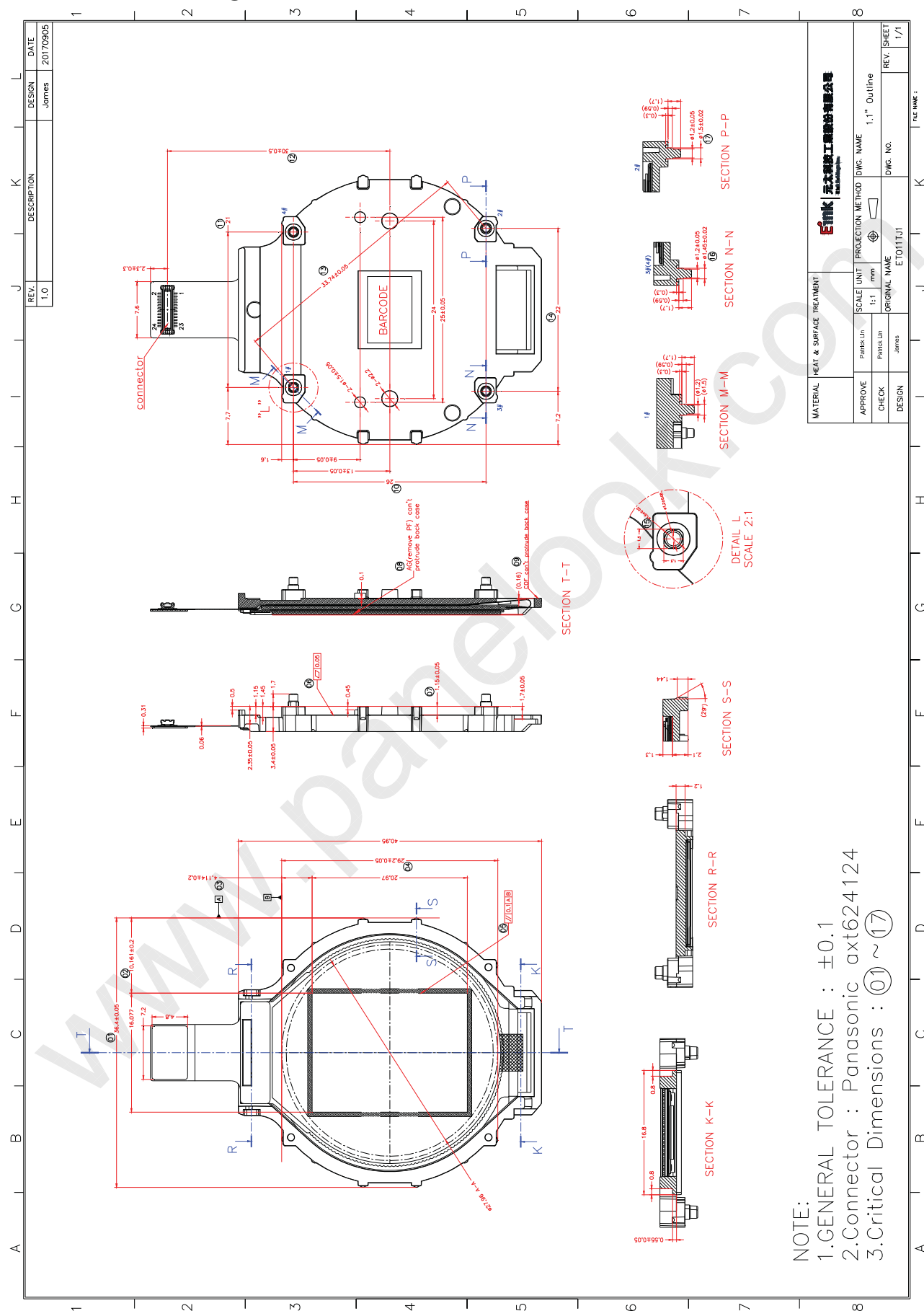
2. Features

- High contrast reflective/electrophoretic technology
- 240x240 display
- High reflectance
- Ultra wide viewing angle
- Pure reflective mode
- Bi-stable
- Commercial temperature range
- Plastic substrate
- With plastic back case

3. Mechanical Specifications

Parameter	Specifications	Unit	Remark
Screen Size	1.1	Inch	
Display Resolution	240 (H) × 240(V)	Pixel	
Active Area	Φ27.96	mm	
Pixel Pitch	116.5(H) × 116.5(V)	μm	
Pixel Configuration	Rectangle		
Outline Dimension	52.8 × 36.4 × 5.1	mm	
Module Weight	2.77±0.3	g	
Display Operating Mode	Reflective mode		

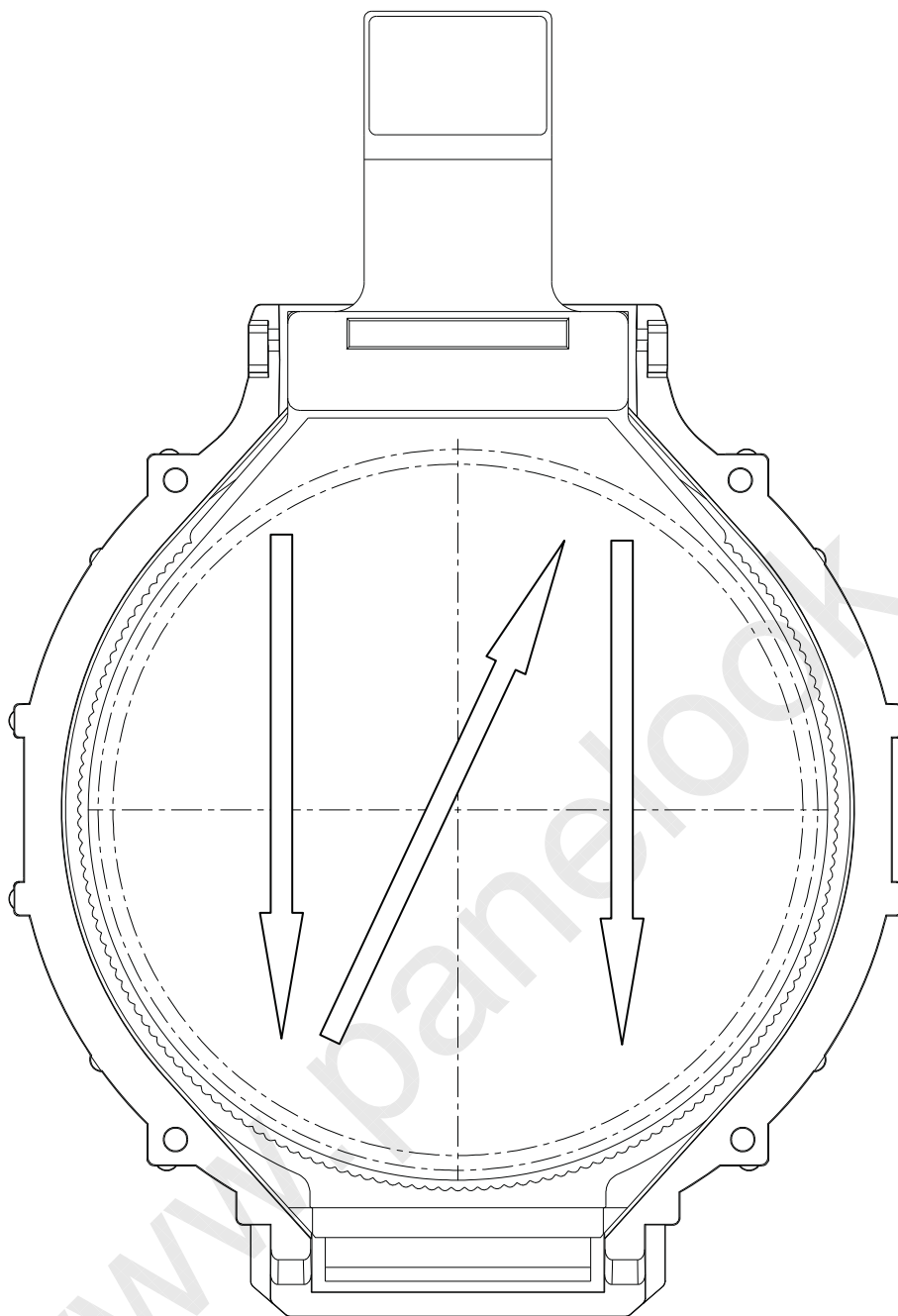
4. Mechanical Drawing of EPD Module



**5. Input /Output Interface****5-1) Connector Type: AXT624124****5-2) Pin Assignment**

Pin #	Single	Description	Remark
1	VCOM	VCOM driving voltage	
2	VGL	Negative Gate driving voltage	
3	VDL	Negative Source driving voltage	
4	VGH	Positive Gate driving voltage	
5	VDH	Positive Source driving voltage	
6	NC	NC	
7	VDDDO	Core logic power pin	
8	VSS	Ground	
9	VDD	Power Supply	
10	SDA	Serial data pin (SPI)	
11	SCL	Serial clock pin (SPI)	
12	CSB	Chip Select input pin	
13	DC	Data /Command control pin	
14	RST_N	Reset	
15	BUSY_N	Busy state output pin	
16	BS	Bus selection pin; L: 4-wire IF. H: 3-wire IF. (Default)	
17	TSDA	I2C Interface to digital temperature sensor Date pin	Note 5-1
18	TSCL	I2C Interface to digital temperature sensor Clock pin	Note 5-1
19	VSHR	Positive driving voltage for driver IC use	
20	RESE	Current Sense Input for the Control Loop	
21	GDR	N-Channel MOSFET Gate Drive Control	
22	VSS	Ground	
23	FL+	LED+(default NC)	
24	FL-	LED-(default NC)	

Note 5-1 : Option for external temperature sensor.

**5-3) Panel Scan Direction**

6. Electrical Characteristics

6-1) Absolute Maximum Ratings:

Parameter	Symbol	Rating	Unit	Remark
Logic Supply Voltage	VDD	-0.3 to +6	V	--
Operating Temp. Range	TOTR	0 to +50	°C	--
Storage Temperature	TSTG	-25 to +60	°C	--

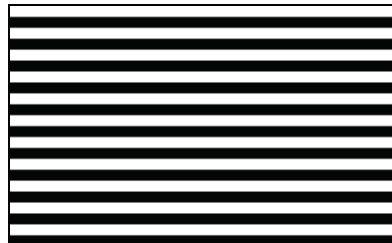
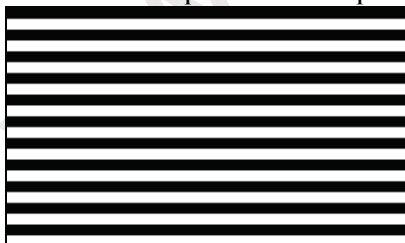
6-2) Panel DC Characteristics

Parameter	Symbol	Conditions	Min	Typ	Max	Unit
Signal ground	VSS		-	0	-	V
Logic voltage supply	VDD		2.3	3.3	3.6	V
	IVDD	VDD=3.3V	-	6.7	18	mA
Common voltage Maximum Power panel	VCOM		-2.76	Adjusted	-1.83	V
Typical power panel Standby power panel	Ptyp		-	22.2	65	mW
	Pstby		-	-	0.1	mW

- The maximum power consumption is measured with following pattern transition: from pattern of repeated 1 consecutive black scan lines followed by 1 consecutive white scan line to that of repeated 1 consecutive white scan lines followed by 1 consecutive black scan lines. (Note 6-1)
- The Typical power consumption is measured with following pattern transition: from horizontal 2 gray scale pattern to vertical 2 gray scale pattern. (Note 6-2)
- The standby power is the consumed power when the panel controller is in standby mode.
-

Note 6-1

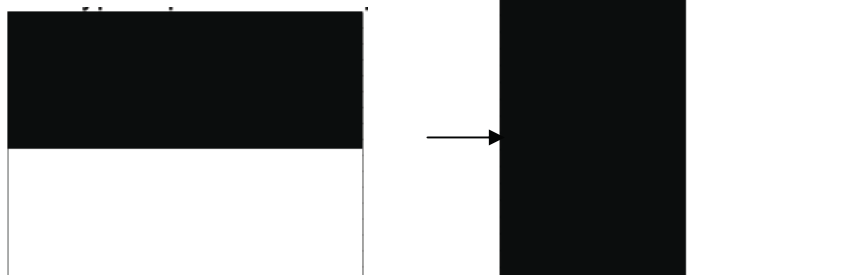
The maximum power consumption

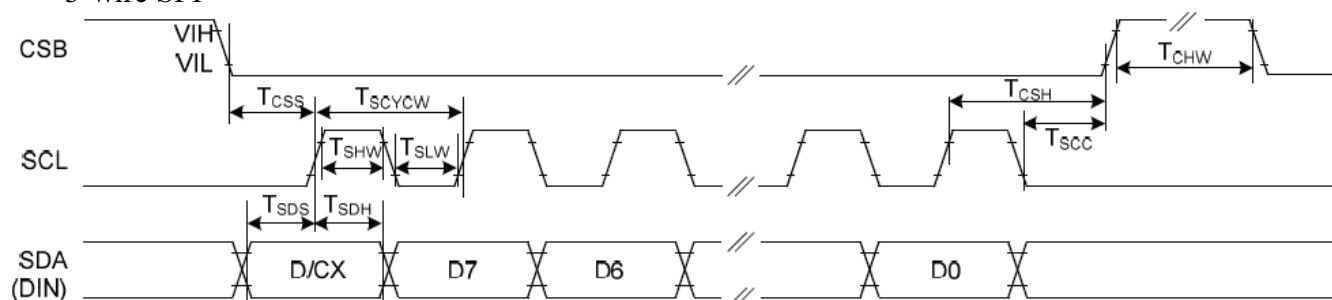
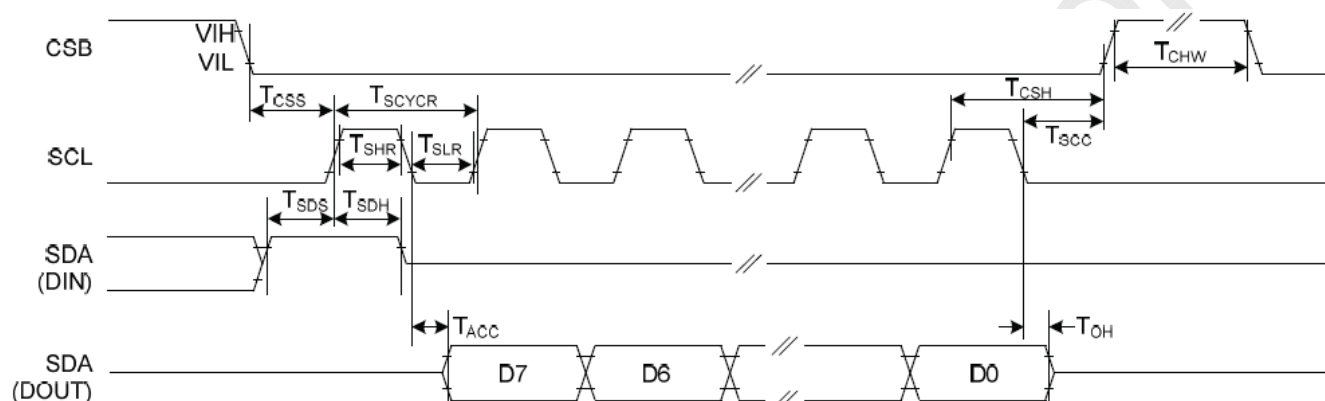




Note 6-2

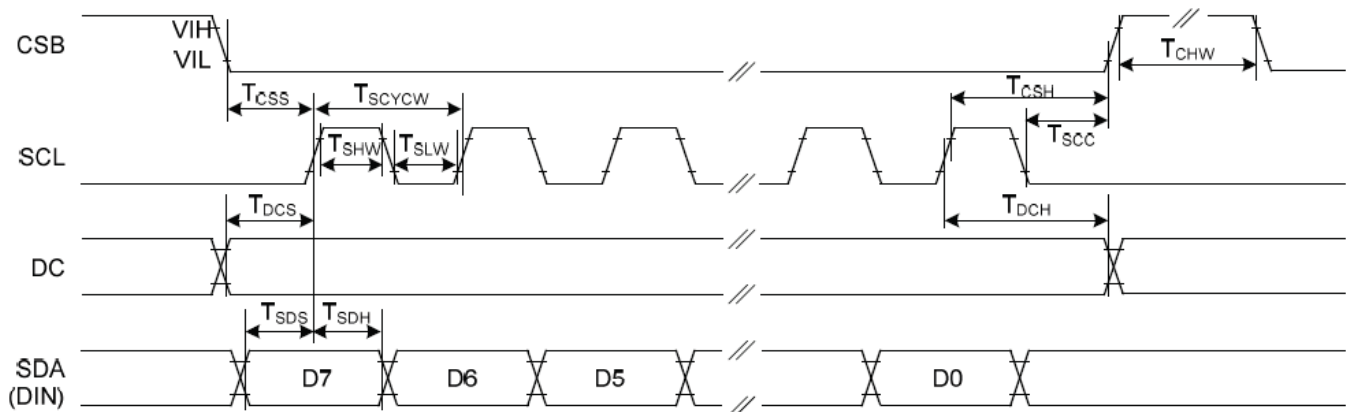
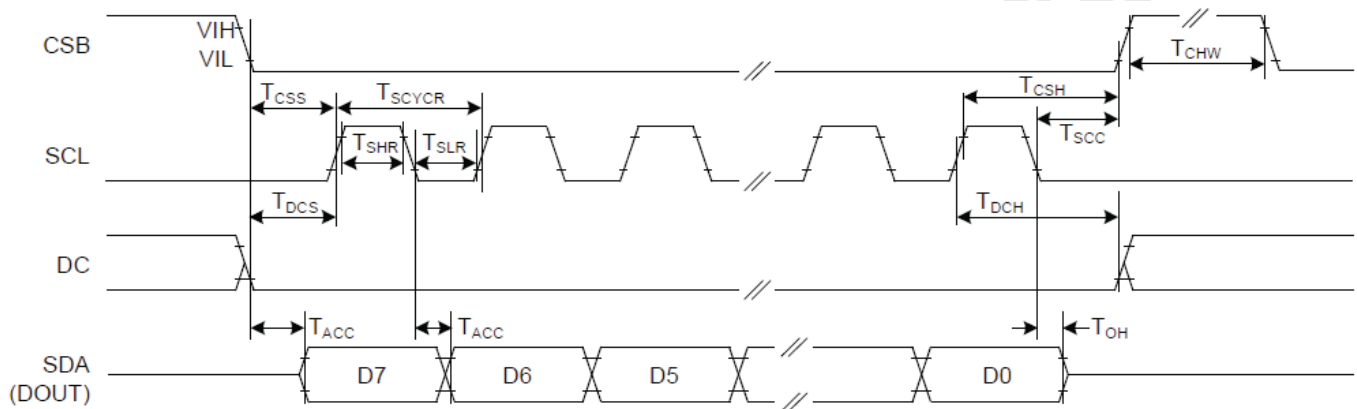
The typical power consumption



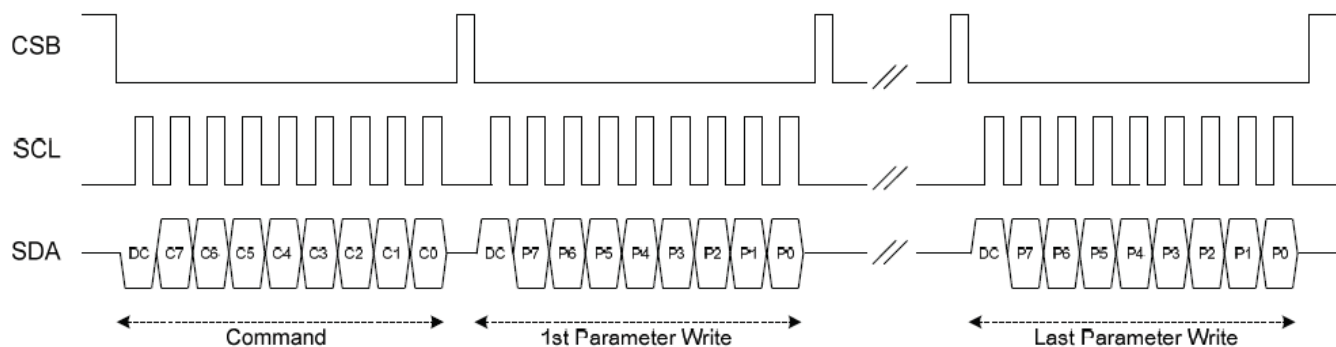
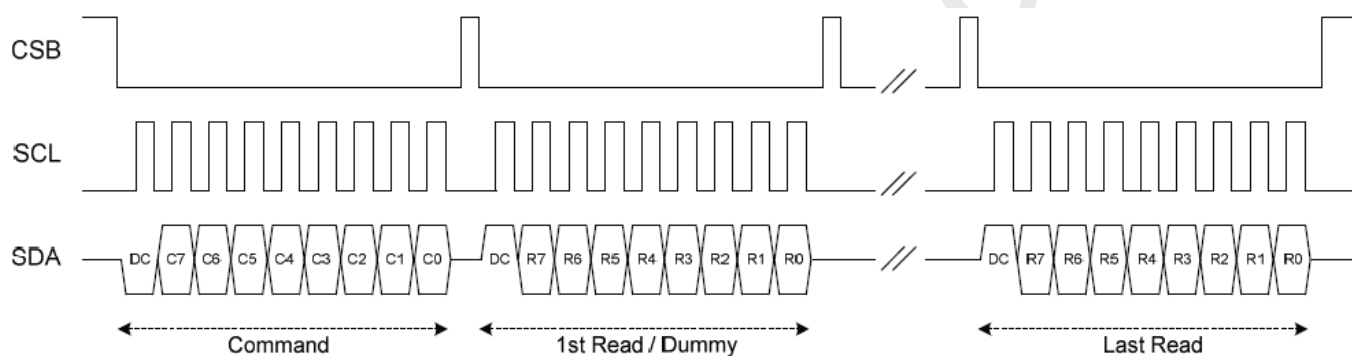
6-3) Panel AC characteristics**3-wire SPI****Figure: 3-wire Serial Interface Characteristics (Write mode)****Figure: 3-wire Serial Interface Characteristics (Read mode)**

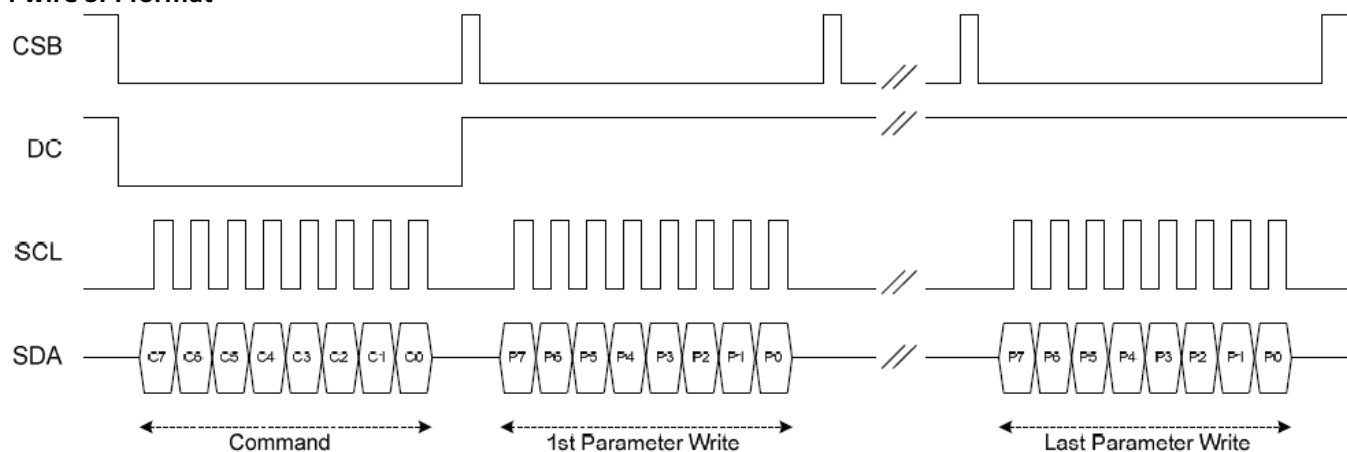
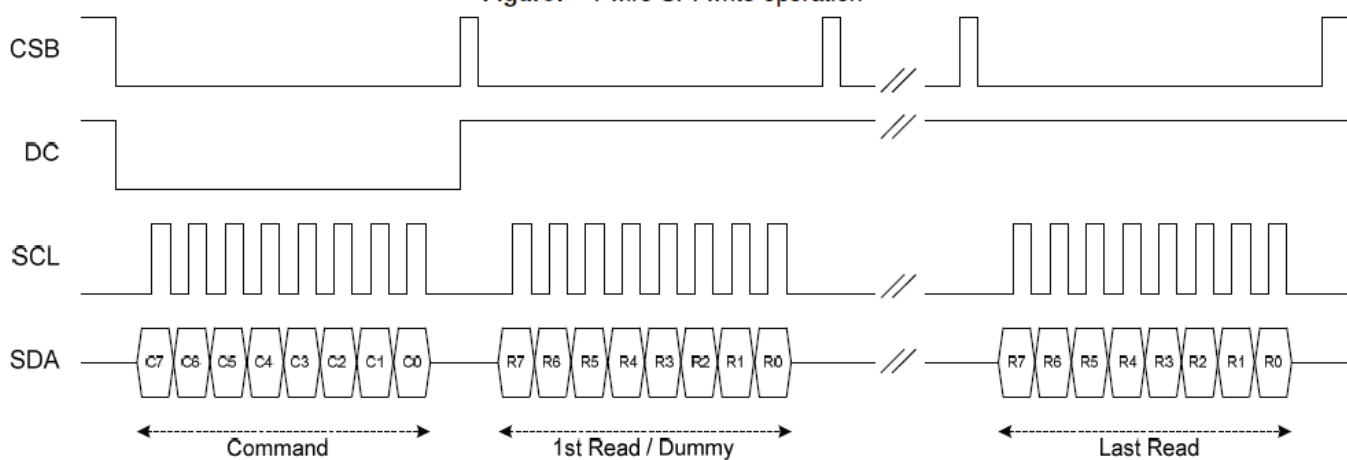
Symbol	Signal / Parameter	Conditions	Min.	Typ.	Max.	Unit
T_{CSS}	CSB	Chip select setup time	60			ns
T_{CSH}		Chip select hold time	65			ns
T_{SCC}		Chip select setup time	20			ns
T_{CHW}		Chip select setup time	40			ns
T_{SCYCW}	SCL	Serial clock cycle (Write)	100			ns
T_{SHW}		SCL "H" pulse width (Write)	35			ns
T_{SLW}		SCL "L" pulse width (Write)	35			ns
T_{SCYCR}		Serial clock cycle (Read)	150			ns
T_{SHR}		SCL "H" pulse width (Read)	60			ns
T_{SLR}		SCL "L" pulse width (Read)	60			ns
T_{SDS}	SDA (DIN)	Data setup time	30			ns
T_{SDH}		Data hold time	30			ns
T_{ACC}	SDA (DOUT)	Access time			50	ns
T_{OH}		Output disable time	15			ns

4-wire SPI

**Figure: 4-wire Serial Interface Characteristics (Write mode)****Figure: 4-wire Serial Interface Characteristics (Read mode)**

Symbol	Signal / Parameter	Conditions	Min.	Typ.	Max.	Unit
T_{CSS}	CSB	Chip select setup time	60			ns
T_{CSH}		Chip select hold time	65			ns
T_{SCC}		Chip select setup time	20			ns
T_{CHW}		Chip select setup time	40			ns
T_{SCYCW}	SCL	Serial clock cycle (Write)	100			ns
T_{SHW}		SCL "H" pulse width (Write)	35			ns
T_{SLW}		SCL "L" pulse width (Write)	35			ns
T_{SCYCR}		Serial clock cycle (Read)	150			ns
T_{SHR}		SCL "H" pulse width (Read)	60			ns
T_{SLR}		SCL "L" pulse width (Read)	60			ns
T_{DCS}	DC	DC setup time	30			ns
T_{DCH}		DC hold time	30			ns
T_{SDS}	SDA (DIN)	Data setup time	30			ns
T_{SDH}		Data hold time	30			ns
T_{ACC}	SDA (DOUT)	Access time			50	ns
T_{OH}		Output disable time	15			ns

6-4) Controller Timing**SPI COMMAND****3 wire SPI format****Figure: 3-wire SPI write operation****Figure: 3-wire SPI read operation**

4 wire SPI format**Figure: 4-wire SPI write operation****Figure: 4-wire SPI read operation**

**6-5) Recommended driving condition for Front Light**

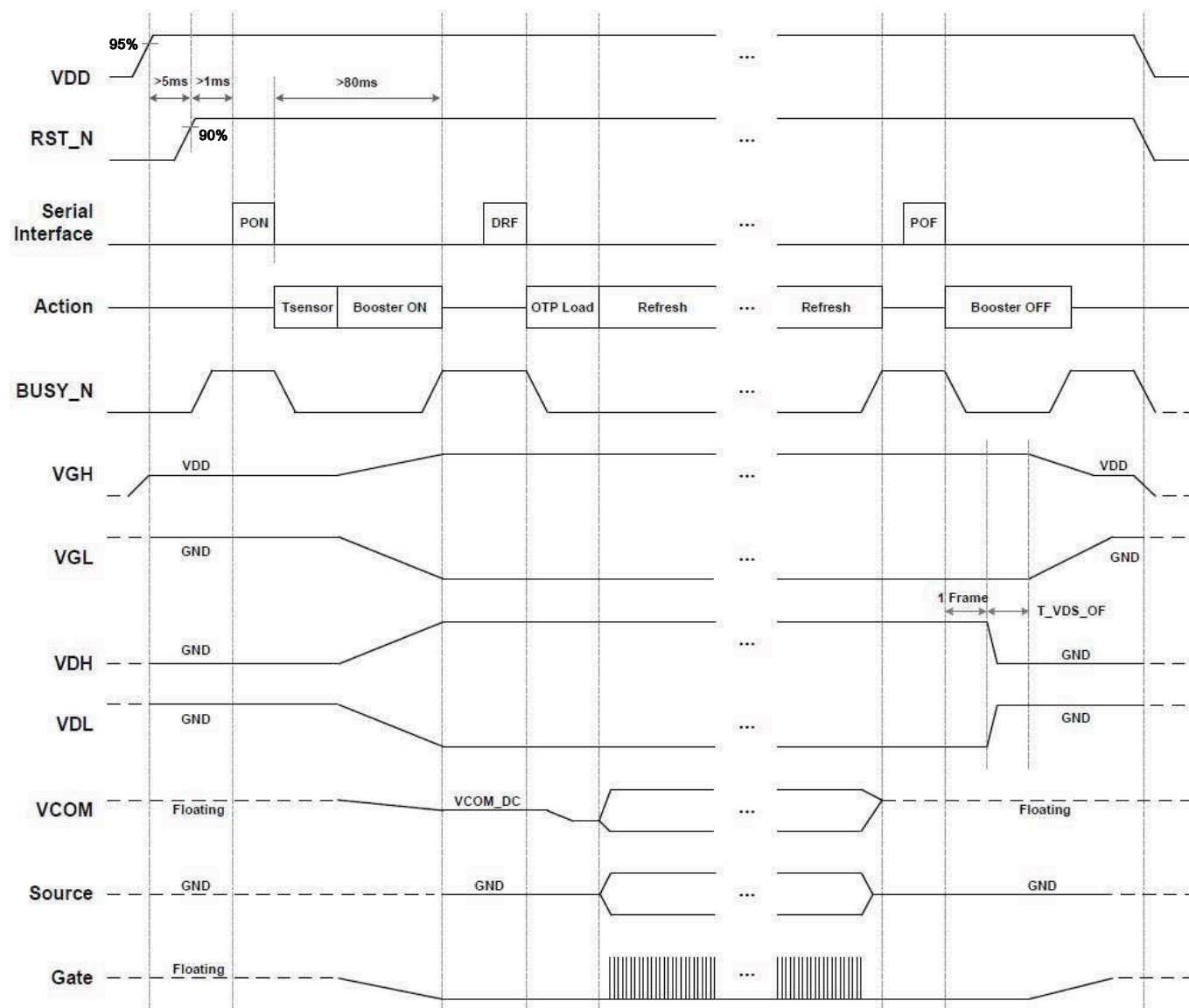
GND = 0 V , Ta = 25°C

Parameter	Symbol	Min	TYP	MAX	Unit
Supply voltage of LED Front light	V _{LED}	-	5.2	5.6	V
Supply current of LED Front light	I _{LED}	-	2	20	mA
Front light Power Consumption	P _{LED}	-	10.4	112	mW
Remark			Note 6-4	Note 6-5	

Note 6-4 : I_{LED}=2 mA with 1-serial LED circuit(2 LEDs)Note 6-5 : I_{LED}=20 mA with 1-serial LED circuit(2 LEDs)

7. Power Sequence

POWER ON/OFF Sequence



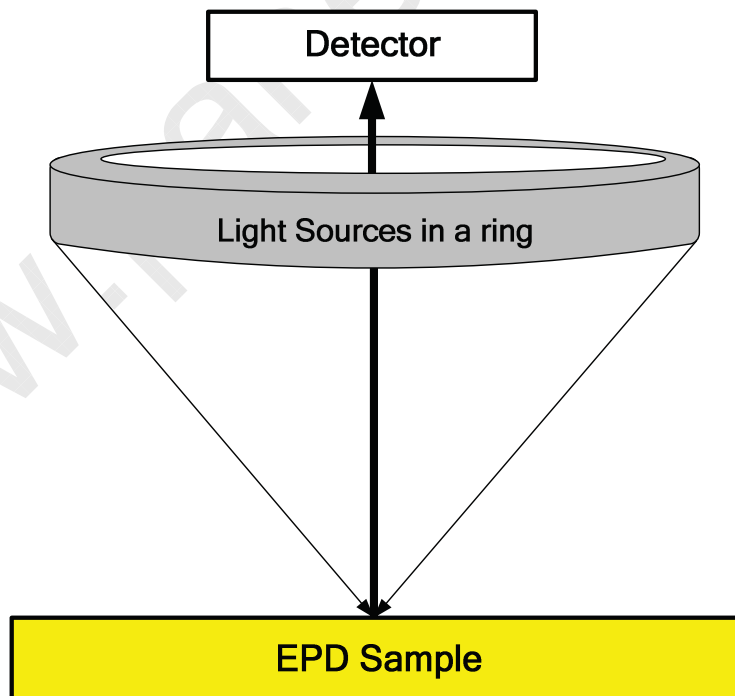
8. Optical Characteristics

8-1) Specifications

Measurements are made by PR655 with MS-75 or equivalent SepctraScan Colorimeter with that the illumination is at an angle 45° from the perpendicular at the center of sample surface, and the detector is perpendicular unless otherwise specified.

T = 25°C

Symbol	Parameter	Conditions		Min	Typ.	Max	Unit	Note
R	Reflectance	White	Front Light off	30	38	-	%	Note 8-1
CR	Contrast Ratio	-		7	11	-	-	Note 8-2
Brightness		$\theta=0^\circ$	Front Light On	-	12	-	cd/m ²	Note 8-3
White Chromaticity		x		0.260	0.300	0.340	-	-
		y		0.250	0.300	0.350	-	
Luminance Uniformity		$\theta=0^\circ$		55	65	-	%	Note 8-4
CR	Contrast Ratio	$\theta=0^\circ$		6	8	-	-	Note 8-5



Note 8-1 : Reflection Ratio

The reflection ratio is expressed as:

$$R = \text{Reflectance Factor}_{\text{white board}} \times (L_{\text{center}} / L_{\text{white board}})$$

L_{center} is the luminance measured at center in a white area (R=G=B=1). $L_{\text{white board}}$ is the

luminance of a standard white board. Both are measured with equivalent illumination source. The viewing angle shall be no more than 2 degrees.

Note 8-2 : Definition of contrast ratio for front light off mode

The contrast ratio (CR) is the ratio between the reflectance in a full white area (R1) and the reflectance in a dark area (Rd): $CR = R1 / Rd$

Note 8-3 : Optical characteristics are determined after the front light has been 'ON' and stable for approximately 60 minutes in a dark environment at 25°C. The values specified are at an approximate distance 50mm from the EPD display surface at a viewing angle of Φ and θ equal to 0°.

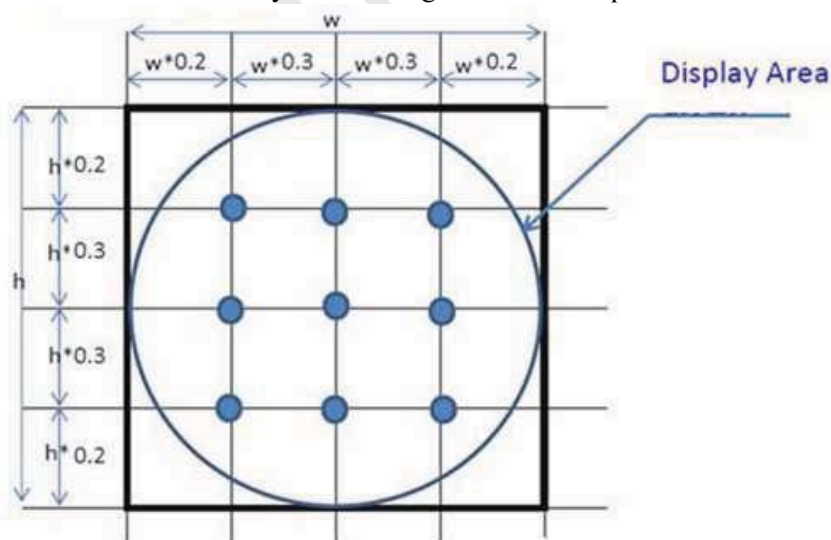
The typical luminance value is measured at LED current 2mA.

Note 8-4 : The uniformity is defined as

$$U = \frac{\text{The Minimum Brightness of the 9 testing Points}}{\text{The Maximum Brightness of the 9 testing Points}}$$

Ambient illumination : < 1 Lux
Measuring direction : Perpendicular to the surface of module
The test pattern is white

The test method of the uniformity is following customer's requirement. Where $h=w=26.8\text{mm}$.



Note 8-5 : Definition of contrast ratio for front light on mode

$$CR = \frac{\text{Luminance when Testing point is White}}{\text{Luminance when Testing point is Black}}$$

**9. Handling, Safety and Environment Requirements and Remark**

WARNING
The display module should be kept flat or fixed to a rigid, curved support with limited bending along the long axis. It should not be used for continual flexing and bending. Handle with care. Should the display break do not touch any material that leaks out. In case of contact with the leaked material then wash with water and soap. Contact E Ink for advice on mounting the display in a curved shape.

CAUTION
The display module should not be exposed to harmful gases, such as acid and alkali gases, which corrode electronics components. Disassembling the display module can cause permanent damage and invalidates the warranty agreements.

Mounting Precautions
(1) It's recommended that you consider the mounting structure so that uneven force (ex. Twisted stress) is not applied to the module.
(2) It's recommended that you attach a transparent protective plate to the surface in order to protect the EPD. Transparent protective plate should have sufficient strength in order to resist external force.
(3) You should adopt radiation structure to satisfy the temperature specification.
(4) Acetic acid type and chlorine type materials for the cover case are not desirable because the former generates corrosive gas of attacking the PS at high temperature and the latter causes circuit break by electro-chemical reaction.
(5) Do not touch, push or rub the exposed PS with glass, tweezers or anything harder than HB pencil lead. And please do not rub with dust clothes with chemical treatment. Do not touch the surface of PS for bare hand or greasy cloth. (Some cosmetics deteriorate the PS)
(6) When the surface becomes dusty, please wipe gently with absorbent cotton or other soft materials like chamois soaks with petroleum benzene. Normal-hexane is recommended for cleaning the adhesives used to attach the PS. Do not use acetone, toluene and alcohol because they cause chemical damage to the PS.
(7) Wipe off saliva or water drops as soon as possible. Their long time contact with PS causes deformations and color fading.



Data sheet status	
Product specification	This data sheet contains preliminary version product specifications.
Limiting values	
Limiting values given are in accordance with the Absolute Maximum Rating System (IEC 134). Stress above one or more of the limiting values may cause permanent damage to the device. These are stress ratings only and operation of the device at these or at any other conditions above those given in the Characteristics sections of the specification is not implied. Exposure to limiting values for extended periods may affect device reliability.	
Application information	
Where application information is given, it is advisory and does not form part of the specification.	
Remark	
All the specifications listed in this document are guaranteed for module only. Post-assembled operation or component(s) may impact module performance or cause unexpected effect or damage and therefore listed specifications is not warranted after any post-assembly operation.	

**10. Reliability Test**

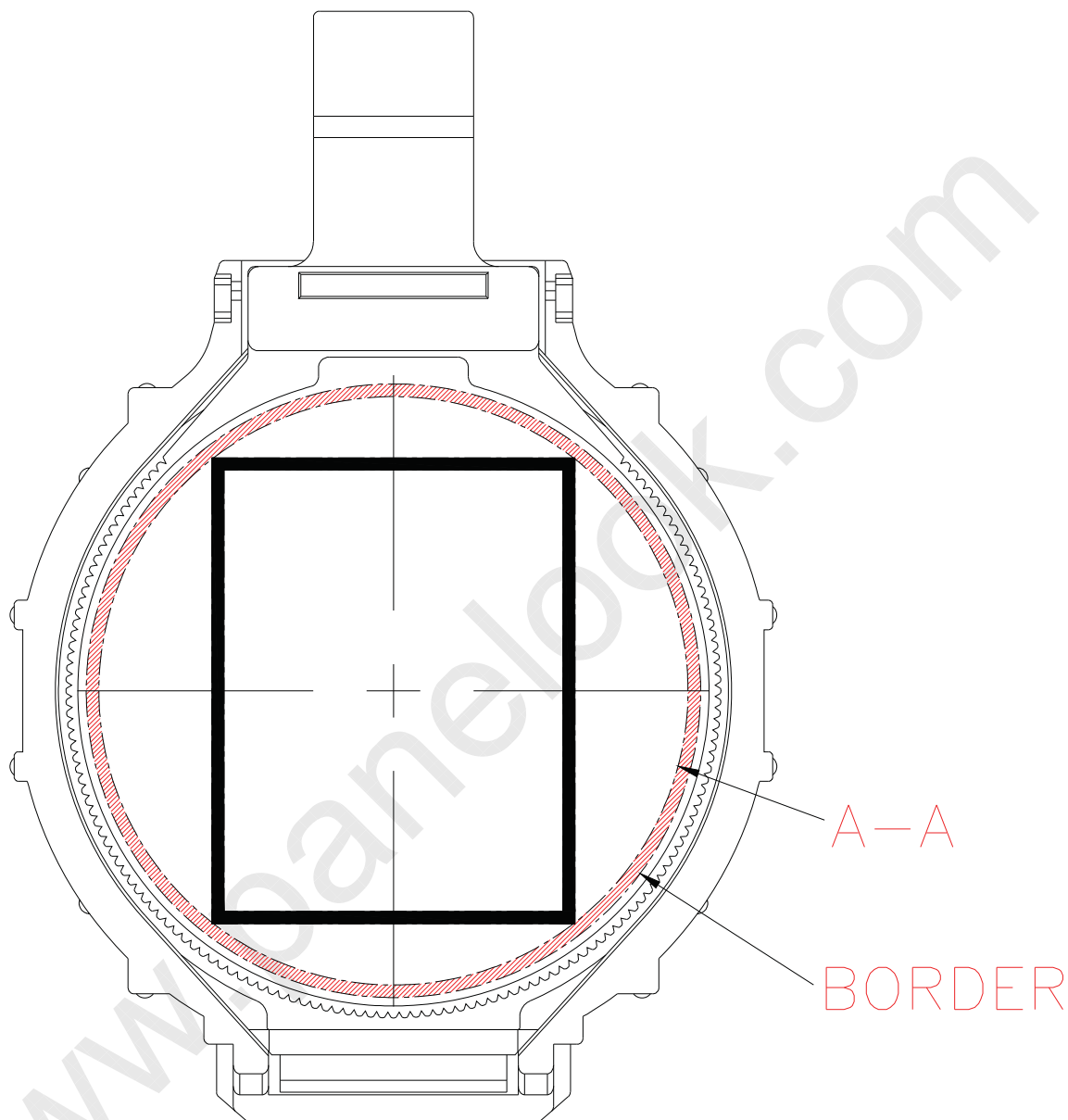
	TEST	CONDITION	METHOD
1	High-Temperature Operation	T = +50°C, RH = 30% , for 240 hrs	IEC 60 068-2-2Bb
2	Low-Temperature Operation	T = 0°C , for 240 hrs	IEC 60 068-2-1Ab
3	High-Temperature Storage	T = +60°C, RH=26% , for 240 hrs (Test In White Pattern)	IEC 60 068-2-2Bb
4	Low-Temperature Storage	T = -25°C , for 240 hrs (Test In White Pattern)	IEC 60 068-2-1Ab
5	High-Temperature, High-Humidity Operation	T = +40°C, RH = 90% , for 168 hrs	IEC 60 068-2-78
6	Temperature Cycle	-25°C→+70°C, 100 Cycles 30min 30min Test in white pattern	IEC 60 068-2-14
7	Solar radiation test	765 W/m ² , 40°C, for 168 hrs (Test In White Pattern)	IEC60 068-2-5Sa
8	Package Vibration	1.04G, Frequency: 10~500Hz Direction: X,Y,Z Duration: 1 hours in each direction	Full packed for shipment
9	Package Drop Impact	Drop from height of 122 cm on concrete surface. Drop sequence: 1 corner, 3 edges, 6 faces One drop for each.	Full packed for shipment
10	Electrostatic Effect (non-operating)	(Machine model)+/- 250V 0Ω, 200pF	IEC 62179, IEC 62180

Actual EMC level to be measured on customer application

Note: The protective film must be removed before temperature test.

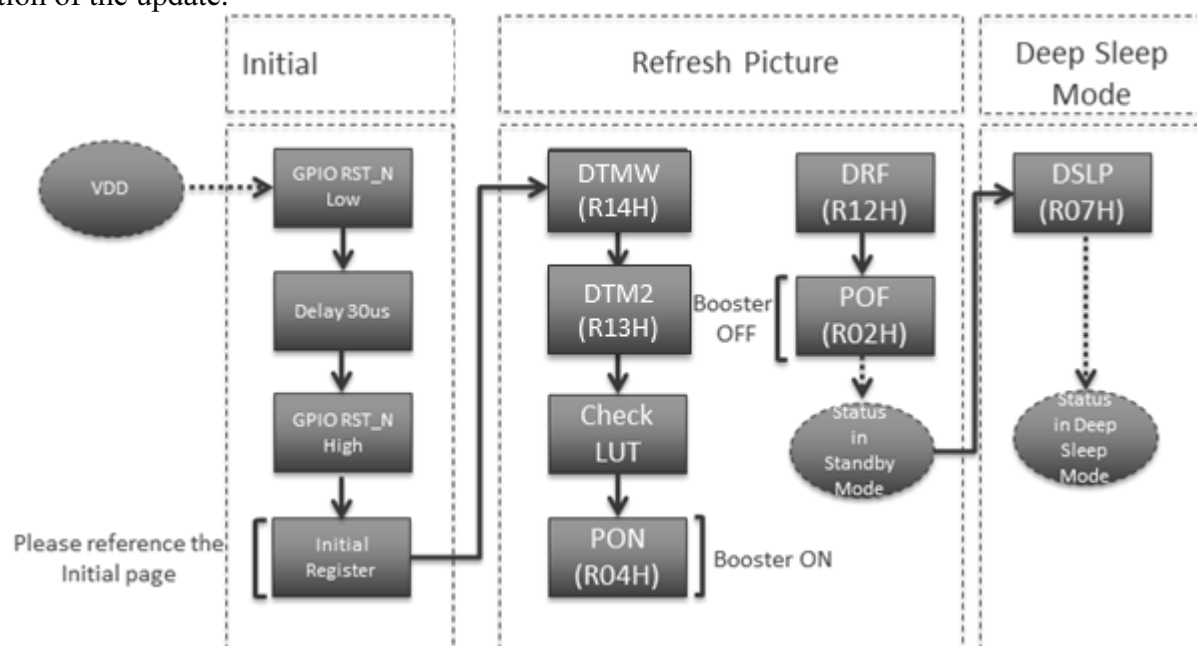
< Criteria >

In the standard conditions, there is not display function NG issue occurred. (including : line defect ,no image). All the cosmetic specification is judged before the reliability stress.

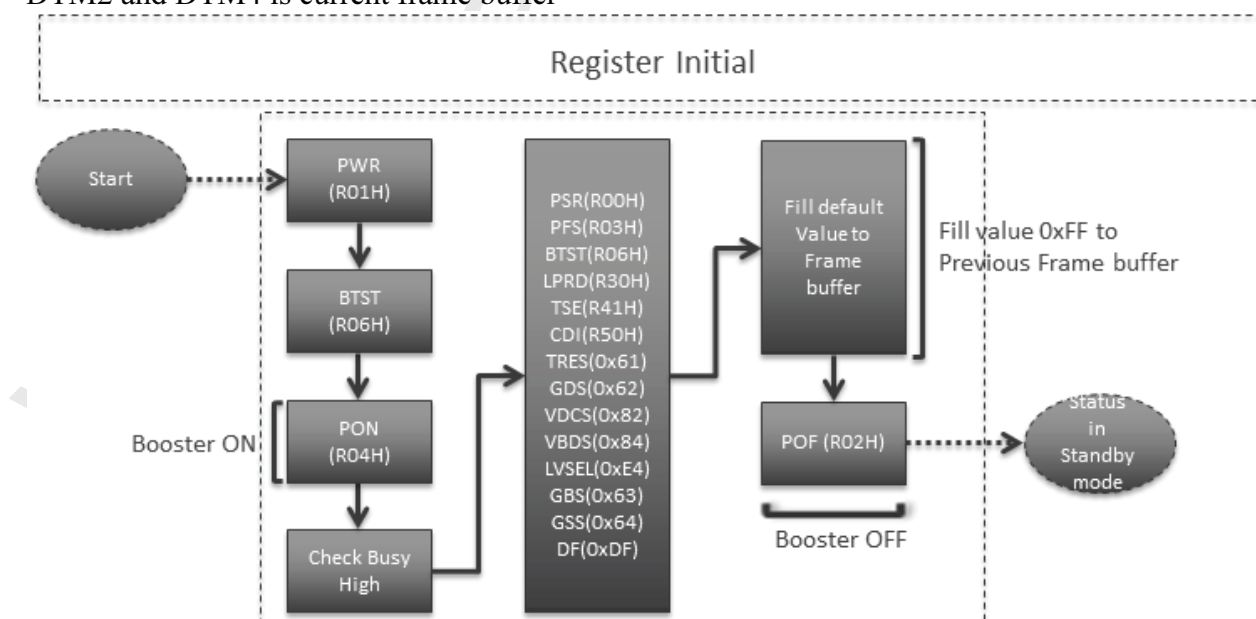
**11. Border Definition**

14. Software Programming Guide

This section describes the image update flow for the ET011TJ1. After the system MCU sends a 2-bit image to the driver IC, the BUSY signal from the panel should be monitored and used to indicate the completion of the update.



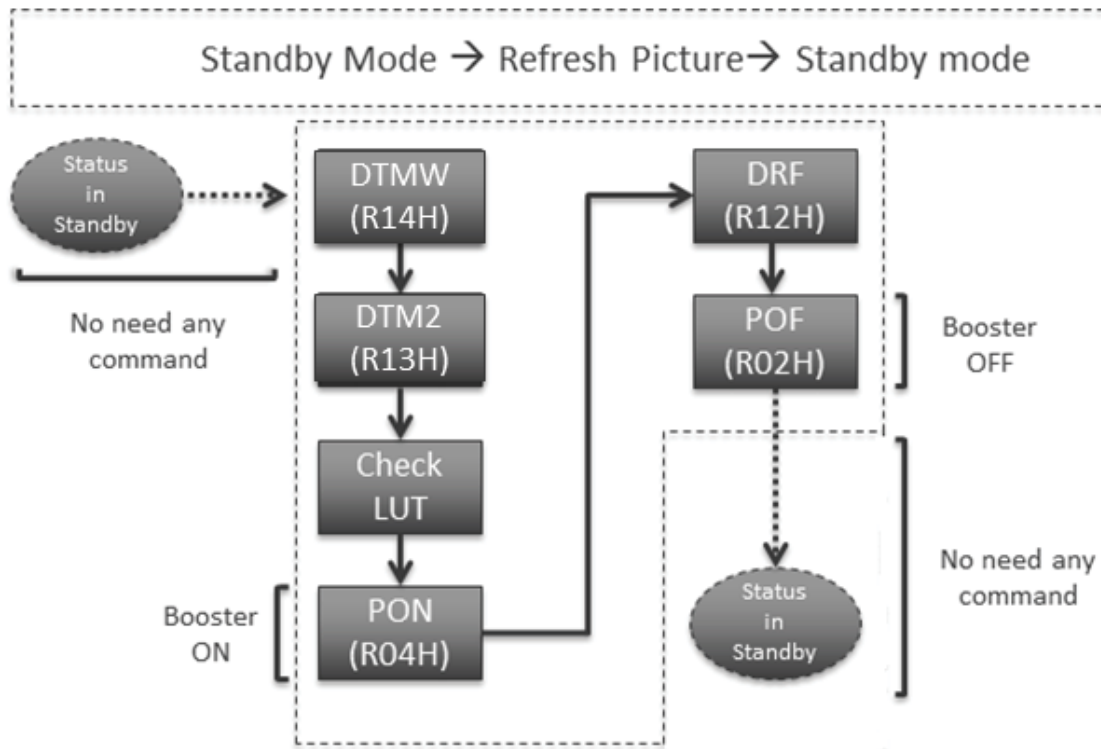
- Please reference Registers initial flow to initial registers
- DTM1 and DTM2 is for 2 Bpp image data
- DTM3 and DTM4 is for 1 Bpp image data
- DTM1 and DTM3 is previous frame buffer
- DTM2 and DTM4 is current frame buffer



- In first time power on, the previous Frame buffer data is random, so we need initial buffer data to white(0xFF is mean white)
- Please reference the sample code to set all registers.

15. User Mode Flow Chart

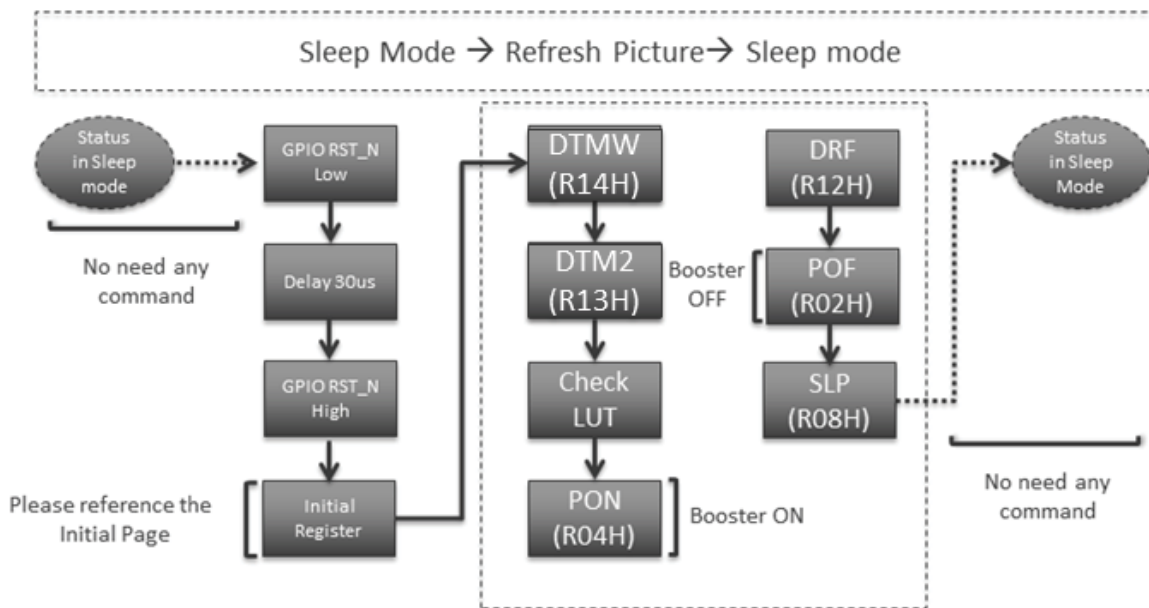
15-1) Standby Mode



Item	Character
VDDDO	1.8V
Booster	OFF
Keep Frame Buffer	Yes
Keep Registers Setting	Yes

Note: Please reference Initial Registers page for initial registers stage

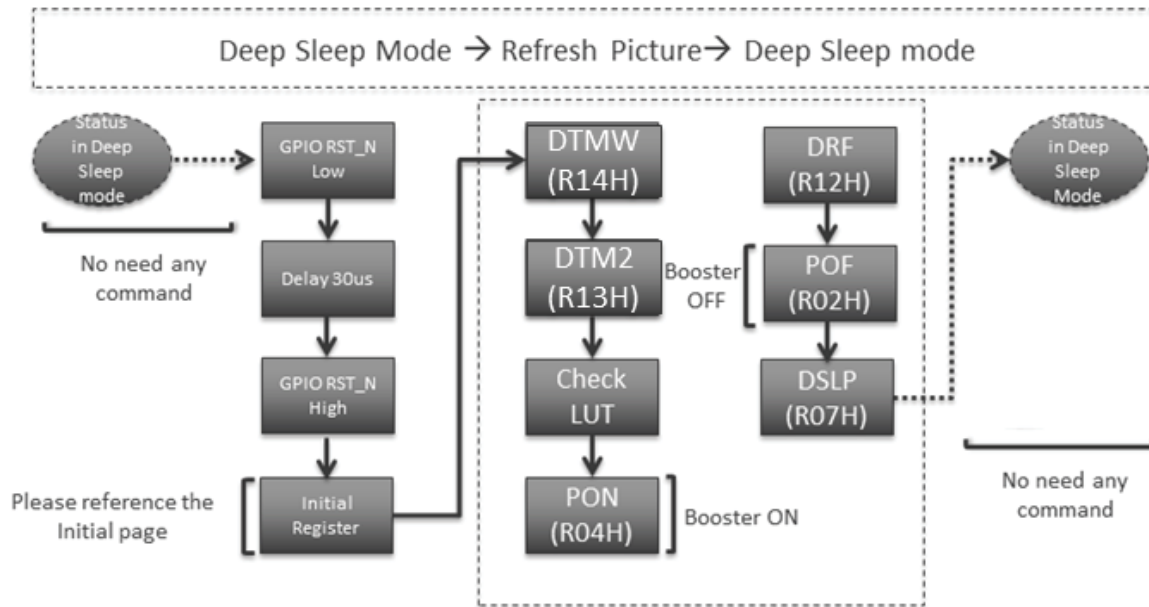
15-2) Sleep Mode



Item	Character
VDDDO	1.0V
Booster	OFF
Keep Frame Buffer	Yes
Keep Registers Setting	Not Guarantee

Note: Please reference Initial Registers page for initial registers stage

15-3) Deep Sleep Mode

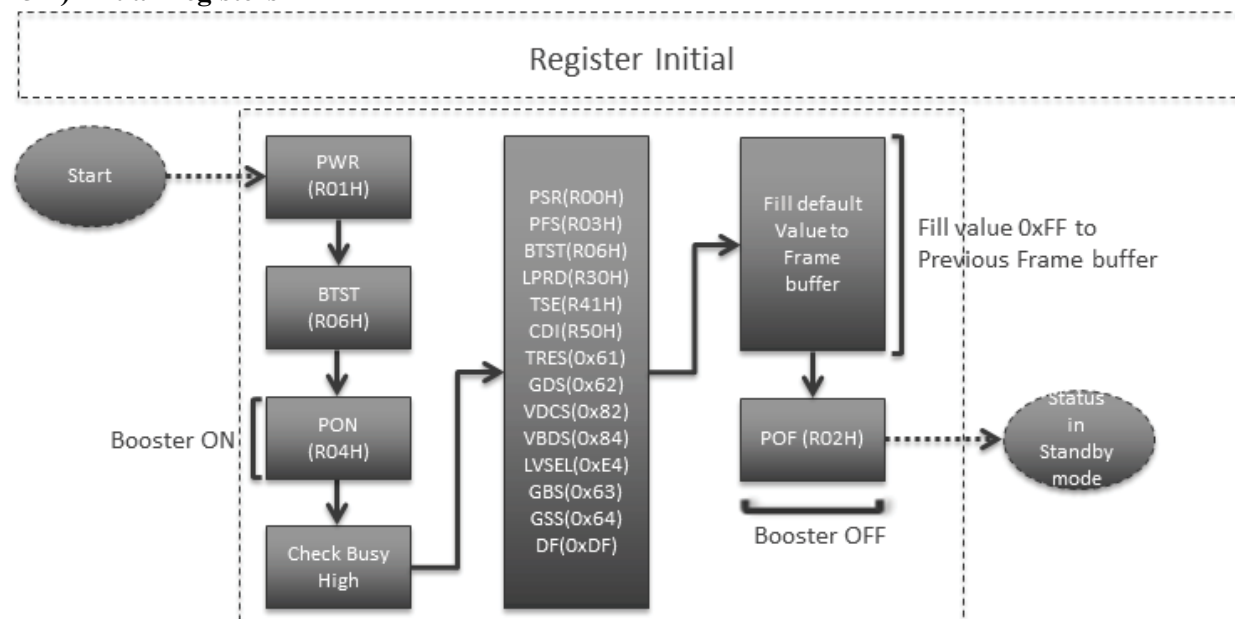


Item	Character
VDDDO	0.0V
Booster	Off
Keep Frame Buffer	No
Keep Registers Setting	No

Note:

1. In “Deep Sleep Mode”, the control signals are recommended tied to 0V and RST_N is released to floating
2. Bare IC Die must be keep away from light which causes photoelectric effect to make internal nodes unstable
3. Please reference Initial Registers page for initial registers stage

15-4) Initial Registers



Note:

1. In first time power on, the previous Frame buffer data is random, so we need initial buffer data to white(0xFF is mean white)
2. Please reference the sample code to set all registers.



16. Command Table

Register Definition

16-1) R00H (PSR): Panel setting Register

Action	W/R	C/D	D7	D6	D5	D4	D3	D2	D1	D0
Setting the panel	W	0	0	0	0	0	0	0	0	0
	W	1	0	0	-	0	1	SHL	SHD_N	RST_N
	W	REG_EN	1	-	-	-	-	1	1	0
	W	1	1	0	0	0	1	0	0	1

NOTE: "-" Don't care, can be set to VDD or GND level

SHL: 0: Shift left.

1: Shift right. (default)

SDMD	SHL	Source order (128 channels)	Source order (Reduced: S63~S60)
0: Edge start	0: Shift right	S127 → ... → S64 → S63 → ... → S0	S127 → ... → S64 → S59 → ... → S0
0: Edge start	1: Shift left	S0 → ... → S63 → S64 → ... → S127	S0 → ... → S59 → S64 → ... → S127
1: Center start	0: Shift right	S63 → ... → S0 → S127 → ... → S64	S59 → ... → S0 → S127 → ... → S64
1: Center start	1: Shift left	S64 → ... → S127 → S0 → ... → S63	S64 → ... → S127 → S0 → ... → S59

SHD_N: Booster Shutdown

0: Booster shutdown and register data kept.

1: No booster shutdown (default)

RST_N: Soft Reset.

0: Assert reset function : Booster OFF and Register data reset to it default value.

1: No effect (default)

REG_EN: LUT selection

0: Using LUT from OTP (default)

1: Using LUT from internal register

**16-2) R01H (PWR): Power setting Register**

Action	W/R	C/D	D7	D6	D5	D4	D3	D2	D1	D0
Selecting Internal/External Power	W	0	0	0	0	0	0	0	0	1
	W	1	-	-	-	-	-	-	VS_EN	VG_EN
	W	1	-	-	-	-	-	0	0	0
	W	1	-	-	0	1	1	1	0	0
	W	1	-	-	0	1	1	1	0	0
	W	1	-	-	0	0	1	0	0	0

NOTE: "-" Don't care, can be set to VDD or GND level

VS_EN: Source power selection

0 : External source power from VDH/VDL pins

1 : Internal DC/DC function for generating VDH/VDL

VG_EN: Gate power selection

0 : External gate power from VGH/VGL pins

1 : Internal DC/DC function for generating VGH/VGL

16-3) R02H (POF): Power OFF Command

Action	W/R	C/D	D7	D6	D5	D4	D3	D2	D1	D0
Turning OFF the power	W	0	0	0	0	0	0	0	1	0

After the Power Off command, driver will power off based on the Power Off Sequence, BUSY_N will become "0". This command will turn off charge pump, T-con, source driver, gate driver, VCOM, and temperature sensor, but register data will be kept until VDD becomes OFF.

SD output and Vcom will base on previous condition. It may have 2 conditions: 0V or floating.

This command can be active only when BUSY_N = "1".

16-4) R03H (PFS): Power off sequence setting Register

Action	W/R	C/D	D7	D6	D5	D4	D3	D2	D1	D0
Setting Power OFF sequence	W	0	0	0	0	0	0	0	1	1
	W	1	-	-	T_VDS_OFF[1:0]	-	-	-	-	-

NOTE: “-” Don’t care, can be set to VDD or GND level

T_VDS_OFF[1:0]: Power OFF Sequence of VDPS and VDNS.

00b: 1 frame (Default)

01b: 2 frames

10b: 3 frames

11b: 4 frame

This command can be active only when BUSY_N = “1”.

16-5) R04H (PON): Power ON Command

Action	W/R	C/D	D7	D6	D5	D4	D3	D2	D1	D0
Turning ON the power	W	0	0	0	0	0	0	1	0	0

After the Power ON command, the driver will be powered ON following the Power ON Sequence. After the Power ON command and all power sequence are ready, the BUSY_N signal will become “1”. Refer to the Power ON Sequence section.

16-6) R06h (BTST): Booster Soft Start

Action	W/R	C/D	D7	D6	D5	D4	D3	D2	D1	D0
Starting data transmission	W	0	0	0	0	0	0	1	1	0
	W	1	BTPH A7	BTPH A6	BTPH A5	BTPH A4	BTPH A3	BTPH A2	BTPH A1	BTPH A0
	W	1	BTPHB 7	BTPHB 6	BTPHB 5	BTPHB 4	BTPHB 3	BTPHB 2	BTPHB 1	BTPHB 0
	W	1	-	-	BTPH A5	BTPH C4	BTPH C3	BTPH C2	BTPH C1	BTPHC 0

NOTE: “-” Don’t care, can be set to VDD or GND level

BTPHA[7:6]: Soft start period of phase A

00b: 10mS

01b: 20mS

10b: 30mS

11b: 40mS

BTPHA[5:3] : Driving strength of phase A

000b: strength 1

001b: strength 2

010b: strength 3

011b: strength 4

100b: strength 5

101b: strength 6

110b: strength 7

111b: strength 8

(strongest)

BTPHA[2:0] : Minimum OFF time setting of GDR in phase B

000b: 0.27uS

001b: 0.34uS

010b: 0.40uS

011b: 0.54uS



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100b: 0.80uS 101b: 1.54uS 110b: 3.34uS 111b: 6.58uS

BTPHB[7:6] : Soft start period of phase B

00b: 10mS 01b: 20mS 10b: 30mS 11b: 40mS

BTPHB[5:3] : Driving strength of phase B

000b: strength 1 001b: strength 2 010b: strength 3 011b: strength 4
 100b: strength 5 101b: strength 6 110b: strength 7 111b: strength 8
 (strongest)

BTPHB[2:0] : Minimum OFF time setting of GDR in phase B

000b: 0.27uS 001b: 0.34uS 010b: 0.40uS 011b: 0.54uS
 100b: 0.80uS 101b: 1.54uS 110b: 3.34uS 111b: 6.58uS

BTPHC[5:3] : Driving strength of phase C

000b: strength 1 001b: strength 2 010b: strength 3 011b: strength 4
 100b: strength 5 101b: strength 6 110b: strength 7 111b: strength 8
 (strongest)

BTPHC[2:0] : Minimum OFF time setting of GDR in phase C

000b: 0.27uS 001b: 0.34uS 010b: 0.40uS 011b: 0.54uS
 100b: 0.80uS 101b: 1.54uS 110b: 3.34uS 111b: 6.58uS

16-7) R07H (DSLP): Deep sleep

Action	W/R	C/D	D7	D6	D5	D4	D3	D2	D1	D0
Deep sleep	W	0	0	0	0	0	0	1	1	1
	W	1	1	0	1	0	0	1	0	1

This command makes the chip enter the deep-sleep mode. The deep sleep mode could return to stand-by mode by hardware reset assertion.

The only one parameter is a check code, the command would be executed if check code is A5h.

16-8) R08H (SLP): sleep

Action	W/R	C/D	D7	D6	D5	D4	D3	D2	D1	D0
sleep	W	0	0	0	0	0	1	0	0	0
	W	1	1	0	1	0	0	1	0	1

This command makes the chip enter the sleep mode. The sleep mode could return to stand-by mode by hardware reset assertion.

The only one parameter is a check code, the command would be executed if check code is A5h.

16-9) R10H (DTM1): Data Start Transmission 1

Action	W/R	C/D	D7	D6	D5	D4	D3	D2	D1	D0
Starting data transmission	W	0	0	0	0	1	0	0	0	0

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	W	1	Pixel0[1:0]	Pixel1[1:0]	Pixel2[1:0]-	Pixel3[1:0]
	W	1	...	...	...	...
	W	1	Pixel(N-4)[1:0]	Pixel(N-3)[1:0]	Pixel(N-2)[1:0]	Pixel(N-1)[1:0]

This command indicates that user starts to transmit data. Then write to old SRAM:

	Source Driver Output	
	DDX=1(Default)	DDX=0
Pixel[1:0]	LUT	LUT
00	Black	White
01	Gray1	Gray2
10	Gray2	Gray1
11	White	Black

16-10) R12H (DRF): Display Refresh

Action	W/R	C/D	D7	D6	D5	D4	D3	D2	D1	D0
(8 byte command)	W	0	0	0	0	1	0	0	1	0
	W	1	-	-	-	PSCA N	RGL_E N	0	Mode[1:0]	
	W	1	X[7:0]							
	W	1	-	-	-	-	-	-	Y[9:8]	
	W	1	Y[7:0]							
	W	1	W[7:0]							
	W	1							L[9:8]	
	W	1	L[7:0]							

PSCAN: Partial Scan control

0: Partial Scan disable.

1: Partial Scan enable (Gate Scan within Display Window only)

RGL_EN: REGAL function control

0: REGAL function disable.

1: REGAL function enable.

MODE: Mode selection, register or OTP.

REG_EN	MODE[1:0]	Description	Mode
0	00	OTP mode-0 LUT	KWG (K/W/Gray)
	01	OTP mode-1 LUT	KW (K/W)
	1x	OTP mode-2 LUT	KWR (K/W/Red)



1	00	Registers (R20h / R22h) KWG LUT and (R26h) FT LUT	KWG (K/W/Gray)
	01		KW (K/W)
	1x	Registers (R21h / R23h / R24h / R25h) Red LUT	KWR (K/W/Red)

X[7:0]: X-axis Start Point. X-axis start point for update display window.

Y[9:0]: Y-axis Start Point. Y-axis start point for update display window.

W[7:0]: X-axis Window Width. X-axis width for update display window.

L[9:0]: Y-axis Window Width. Y-axis width for update display window.

16-11) R13H (DTM2): Data Start Transmission 2

Action	W/R	C/D	D7	D6	D5	D4	D3	D2	D1	D0
Starting data transmission	W	0	0	0	0	1	0	0	1	1
	W	1	Pixel0[1:0]		Pixel1[1:0]		Pixel2[1:0]-		Pixel3[1:0]	
	W	1	...		...		...		...	
	W	1	Pixel(N-4)[1:0]		Pixel(N-3)[1:0]		Pixel(N-2)[1:0]		Pixel(N-1)[1:0]	

This command indicates that user starts to transmit data. Then write to new SRAM:

	Source Driver Output	
	DDX=1(Default)	DDX=0
Pixel[1:0]	LUT	LUT
00	Black	White
01	Gray1	Gray2
10	Gray2	Gray1
11	White	Black

**16-12) R14H (DTMW): Data Start Transmission Window Register**

Action	W/R	C/D	D7	D6	D5	D4	D3	D2	D1	D0
(8 byte command)	W	0	0	0	0	1	0	1	0	0
	W	1	X[7:0]							
	W	1	-	-	-	-	-	-	Y[9:8]	
	W	1	Y[7:0]							
	W	1	W[7:0]							
	W	1							L[9:8]	
	W	1	L[7:0]							

This command indicates the window before user start to transmit data.

The window is defined by (X, Y, W, L). X and W should be 4n format where n is integer.

16-13) R15H (DTM3): Data Start Transmission 3

Action	W/R	C/D	D7	D6	D5	D4	D3	D2	D1	D0
Starting data transmission	W	0	0	0	0	1	0	1	0	1
	W	1	Pixel 0	Pixel 1	Pixel 2	Pixel 3	Pixel 4	Pixel 5	Pixel 6	Pixel 7
	W	1	...	...	...	...	...	...	...	...
	W	1	Pixel(N -8)	Pixel(N -7)	Pixel(N -6)	Pixel(N -5)	Pixel(N -4)	Pixel(N -3)	Pixel(N -2)	Pixel(N -1)

This command transmitting old data to SRAM using KW mode only.

Pixel[0]: data bit for KW,

0: Black

1: White

Source Driver (KW Mode)		
	DDX=1(Default)	DDX=0
Data	LUT	LUT
0	Black	White
1	White	Black

**16-14) R16H (DTM4): Data Start Transmission 4**

Action	W/R	C/D	D7	D6	D5	D4	D3	D2	D1	D0
Starting data transmission	W	0	0	0	0	1	0	1	1	0
	W	1	Pixel 0	Pixel 1	Pixel 2	Pixel 3	Pixel 4	Pixel 5	Pixel 6	Pixel 7
	W	1	...	...	...	...	...	...	...	...
	W	1	Pixel(N-8)	Pixel(N-7)	Pixel(N-6)	Pixel(N-5)	Pixel(N-4)	Pixel(N-3)	Pixel(N-2)	Pixel(N-1)

This command transmitting old data to SRAM using KW mode only.

Pixel[0]: data bit for KW,

0: Black

1: White

Source Driver (KW Mode)		
	DDX=1 (Default)	DDX=0
Data	LUT	LUT
0	Black	White
1	White	Black

16-15) R30H (LPRD): PLL Control

Action	W/R	C/D	D7	D6	D5	D4	D3	D2	D1	D0
Controlling PLL	W	0	0	0	1	1	0	0	0	0
	W	1	LCLK[7:0]							

The command controls the PLL clock frequency. The PLL structure supports the following frame rates:

(FR: Frame Rate, Unit: Hz)

LCLK[7:0]: Line Period Setting. Specify clock cycle number of 1MHz for one line period. (LCLK >=4)

04h: 5 clock cycles

12h: 19 clock cycles

**16-16) R40H (TSC) Temperature Sensor Command**

Action	W/R	C/D	D7	D6	D5	D4	D3	D2	D1	D0
Sensing Temperature	W	0	0	1	0	0	0	0	0	0
	R	1	D[10:3] or TS[7:0]							
	R	1	D[2:0]			-	-	-	-	-

This command reads the temperature sensed by the temperature sensor.

TS[7:0]: When TSE (R41h) is set to 0, this command reads internal temperature sensor value.

D[10:0]: When TSE (R41h) is set to 1, this command reads external LM75 temperature sensor value.

TS[7:0]	Temperature(°C)
1100 1110b	-25
1100 1111b	-24.5
1101 0000b	-24.5
:	:
1111 1110b	-1
1111 1111b	-0.5
0000 0000b	0
0000 0001b	0.5
0000 0010b	1
:	:
0101 1010b	45
:	:
0110 0011b	49.5
0110 0100b	50

**16-17) R41H (TSE) Temperature Sensor Enable**

Action	W/R	C/D	D7	D6	D5	D4	D3	D2	D1	D0
Calibrate Temperature Sensor	W	0	0	1	0	0	0	0	0	1
	W	1	TSE	-	-	-	TO[3:0]			

This command selects Internal or External temperature sensor.

TSE: Internal temperature sensor switch

0: Enable (default)

1: Disable; using external sensor.

TO[3:0]: Temperature Offset

TO[3:0]	Temperature Offset	TO[3:0]	Temperature Offset
0000	+0(Default)	1000	-8.0
0001	+1.0	1001	-7.0
0010	+2.0	1010	-6.0
0011	+3.0	1011	-5.0
0100	+4.0	1100	-4.0
0101	+5.0	1101	-3.0
0110	+6.0	1110	-2.0
0111	+7.0	1111	-1.0

16-18) R42H (TSW) Temperature Sensor Write

Action	W/R	C/D	D7	D6	D5	D4	D3	D2	D1	D0
Calibrate Temperature Sensor	W	0	0	1	0	0	0	0	1	0
	W	1	WATTR[7:0]							
	W	1	WMSB[7:0]							
	W	1	WLSB[7:0]							

This command reads the temperature sensed by the temperature sensor.

WATTR: D[7:6]: I²C Write Byte Number

00 : 1 byte (head byte only)

01 : 2 bytes (head byte + pointer)

10 : 3 bytes (head byte + pointer + 1st parameter)

11 : 4 bytes (head byte + pointer + 1st parameter + 2nd parameter)

D[5:3]: User-defined address bits (A2, A1, A0)

D[2:0]: Pointer setting

WMSB[7:0]: MSByte of write-data to external temperature sensor

WLSB[7:0]: LSByte of write-data to external temperature sensor

16-19) R43H (TSR) Temperature Sensor Read

Action	W/R	C/D	D7	D6	D5	D4	D3	D2	D1	D0
Calibrate Temperature Sensor	W	0	0	1	0	0	0	0	1	1
	W	1	RMSB[7:0]							
	W	1	RLSB[7:0]							

This command reads the temperature sensed by the temperature sensor.

RMSB[7:0]: MSByte read data from external temperature sensor

RLSB[7:0]: LSByte read data from external temperature sensor

16-20) R50H (CDI) VCOM and Data interval setting

Action	W/R	C/D	D7	D6	D5	D4	D3	D2	D1	D0
CDI	W	0	0	1	0	1	0	0	0	0
	W	1	BDZ	BDV	VBD[1:0]		-	-	-	DDX
	W	1	DCI[3:0]				-	-	CDI[9:8]	
	W	1	CDI[7:0]							

This command indicates the interval of Vcom and data output. When setting the vertical back porch, the total blanking will be kept (20 Hsync).

BDZ: Border Hi-Z control

0: Border Output Hi-Z disabled. (default)

1: Border Output Hi-Z enabled.

BDV: Border DC Voltage control

0: Border Output DC Voltage Function disabled. (default)

1: Border Output DC Voltage Function enabled.

VBD[1:0]: Border output selection

DDX[1:0]: Data polarity.

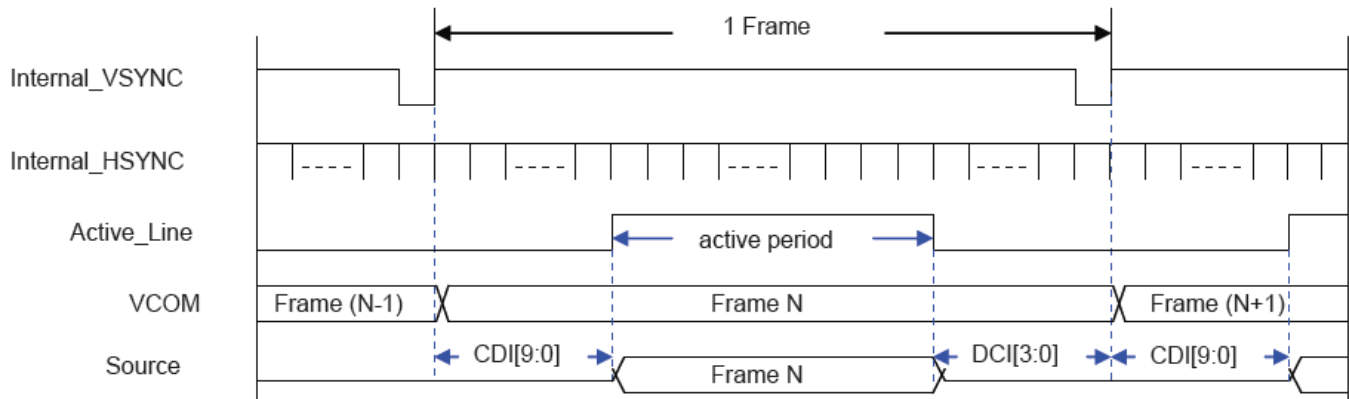
	Border Output	
	DDX=1(Default)	DDX=0
VBD[1:0]	LUT	LUT
00	Black	White
01	Black	White
10	White	Black
11	White	Black

DCI[3:0]: Source to VCOM interval. Interval time setting from source data to VCOM.

0000b ~ 1111b: 1 Hsync ~ 16 Hsync, respectively. **(Default: 001b: 4 Hsync)**

CDI[9:0]: VCOM to Source interval. Interval time setting from VCOM to source dat.

000 0000 000b ~ 11 1111 1111b: 1 Hsync ~ 1023 Hsync, respectively. **(Default: 018h: 25 Hsync)**



16-21) R51H (LPD) Low Power Detection

Action	W/R	C/D	D7	D6	D5	D4	D3	D2	D1	D0
Detect Low Power	W	0	0	1	0	1	0	0	0	1
	W	1	-	-	-	-	-	-	-	LPD

This command indicates the input power condition. Host can read this flag to learn the battery condition.

LPD: Internal temperature sensor switch

0: Low power input ($V_{DD} < 2.2V/2.3V/2.4V/2.5V$) 1: Normal status (default)

16-22) R61H (TRES) Resolution Setting

Action	W/R	C/D	D7	D6	D5	D4	D3	D2	D1	D0
Set Display Resolution	W	0	0	1	1	0	0	0	0	1
	W	1	1	1	1	0	1	1	1	1
	W	1	0	0	0	0	0-	0	0	0
	W	1	1	1	1	0	1	1	1	1

**16-23) R70H (REV) Revision**

Action	W/R	C/D	D7	D6	D5	D4	D3	D2	D1	D0
Chip Revision	W	0	0	1	1	1	0	0	0	0
	W	1	LUTVER0[7:0]							
	W	1	LUTVER1[7:0]							

LUTVER0[7:0]: LUT version 0 (Located in internal OTP address: F81h)

LUTVER1[7:0]: LUT version 1 (Located in internal OTP address: F82h)

This command only active when BUSY_N="1".

16-24) R71H (FLG) Get Status

Action	W/R	C/D	D7	D6	D5	D4	D3	D2	D1	D0
Read Flags	W	0	0	1	1	1	0	0	0	1
	W	1	-	-	I ² C_ER R	I ² C_ BUSY N	data_ flag	PON	POF	BUSY_ N

This command reads the IC status.

I²C_ERR: I²C master error status

I²C_BUSYN: I²C master busy status (low active)

data_flag: Driver has already received all the one frame data

PON: Power ON status

POF: Power OFF status

BUSY_N: Driver busy status (low active)

**16-25) R81H (VV) VCOM value**

Action	W/R	C/D	D7	D6	D5	D4	D3	D2	D1	D0
Automatically measure Vcom	R	0	1	0	0	0	0	0	0	1
	R	1	-	VV[6:0]						

This command gets the Vcom value.

VV[6:0]: Vcom Value

VV[6:0]	VCOM Value
000 0000b	-0.1V
000 0001b	-0.15V
000 0010b	-0.20V
000 0011b	-0.25V
000 0100b	-0.30V
:	:
100 1110b	-4.0V
(others)	-4.0V

**16-26) R82H (VDCS) VCM_DC Setting**

Action	W/R	C/D	D7	D6	D5	D4	D3	D2	D1	D0
Set VCM_DC	W	0	1	0	0	0	0	0	1	0
	W	1	-	VDCS[60]						

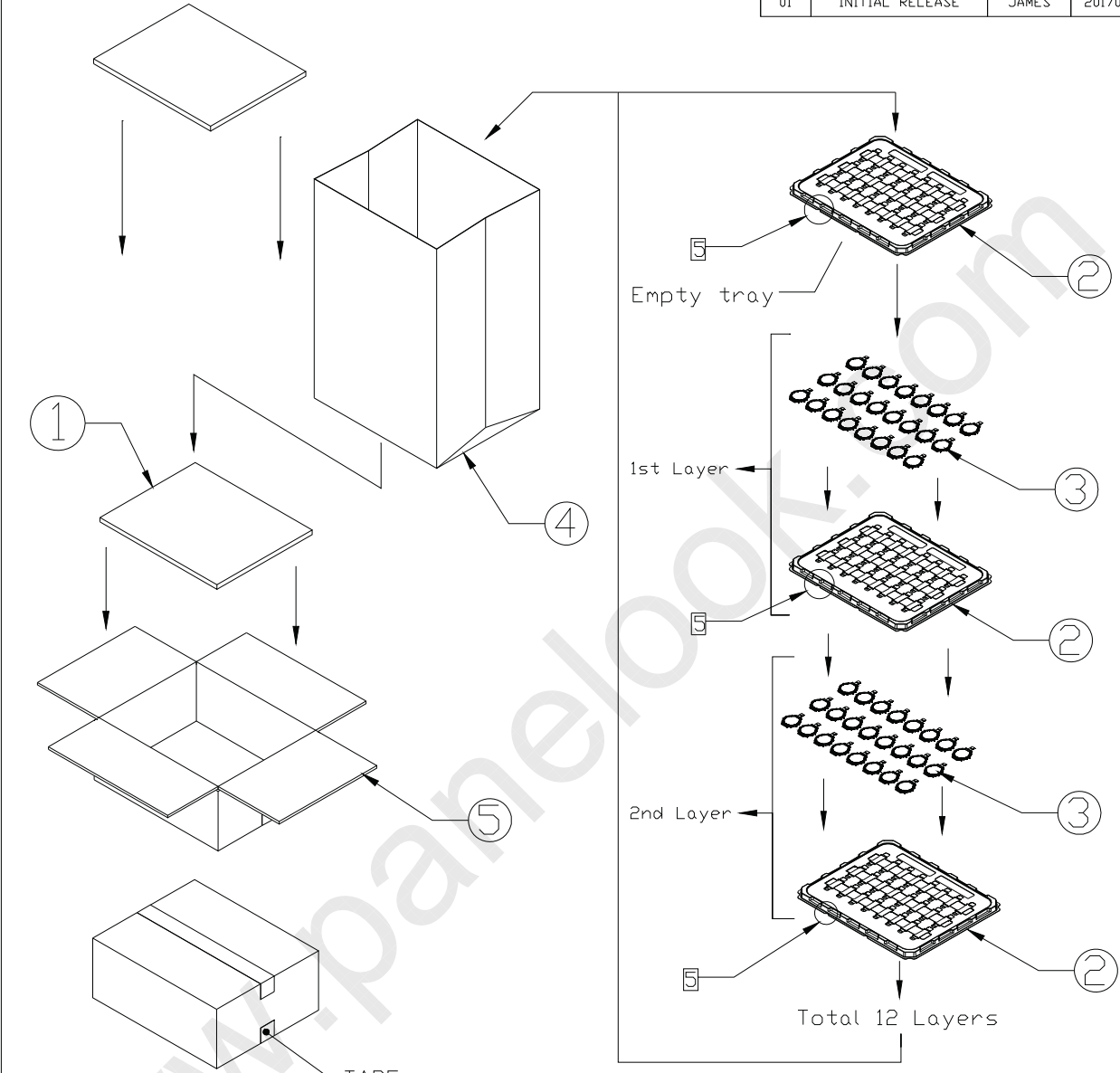
This command sets VCOM_DC value

VDCS[6:0]: Vcom_DC setting

VDCS[6:0]	VCOM_DC value
000 0000b	-0.1V
000 0001b	-0.15V
000 0010b	-0.20V
000 0011b	-0.25V
000 0100b	-0.30V
:	:
100 1110b	-4.0V
(others)	-4.0V

17. Packing

REV.	DESCRIPTION	DESIGN	DATE
01	INITIAL RELEASE	JAMES	20170313



NOTE:

- ONE LAYER INCLUDE:
24 PCS MODULE & 1 PIECE OF TRAY.
- Q'TY:288 PCS PANEL/CARTON.
- DIMENSION: 455*375*190mm.
- WEIGHT: 4.0 Kg.
- MAKE SURE TRAY STACKED WITH 0° ROTATION.
WE CAN CHECK THIS BY LATERAL SIDE VIEW.

ITEM	DESCRIPTION	Q'ty	REMARK
5	CARTON INTERNAL	1	
4	摺口袋 450*380*700mm	1	抗靜電
3	ET011TJ1	288	
2	TRAY	13	抗靜電
1	EPE FOAM	2	

MTL.SPEC.		UNSPECIFIED TOL.		REMARK	
		ANGLE			
		ROUGHNESS			
APPROVE	PATRICK LIN	SCALE	UNIT	SHEET	DWG.TITLE
CHECK	PATRICK LIN			1 OF 1	1.1" PACKING DRAW.
DESIGN	JAMES	MTL.NO.		ET011TJ1	DWG.NO.
					REV. 01
					A4 SIZE

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